

12.1 INTRODUCTION

In general, the term 'game' refers to a situation of conflict and competition in which two or more competitors (or participants) are involved in the decision-making process in anticipation of certain outcomes over a period of time. The competitors are referred to as *players*. A player may be an *individual*, *individuals*, or an *organization*. A few examples of competitive and conflicting decision environment, that involve the interaction between two or more competitors are:

- Pricing of products, where sale of any product is determined not only by its price but also by the price set by competitors for a similar product
- The success of any TV channel programme largely depends on what the competitors presence in the same time slot and the programme they are telecasting.
- The success of a business strategy depends on the policy of internal revenue service regarding the expenses that may be disallowed,
- The success of an advertising/marketing campaign depends on various types of services offered to the customers.

For academic interest, *theory of games* provides a series of mathematical models that may be useful in explaining interactive decision-making concepts, where two or more competitors are involved under conditions of conflict and competition. However, such models provide an opportunity to a competitor to evaluate not only his personal decision alternatives (courses of action), but also the evaluation of the competitor's possible choices in order to win the game.

Game theory came into existence in 20th Century. However, in 1944 John Von Neumann and Oscar Morgenstern published a book named *Theory of Games and Economic Behavior*, in which they discussed how businesses of all types may use this technique to determine the best strategies given a competitive business environment. The author's approach was based on the principle of '*best out of the worst*'.

The models in the *theory of games* can be classified based on the following factors:

Number of players If a game involves only two players (competitors), then it is called a *two-person game*. However, if the number of players are more, the game is referred to as *n-person game*.

Sum of gains and losses If, in a game, the sum of the gains to one player is exactly equal to the sum of losses to another player, so that, the sum of the gains and losses equals zero, then the game is said to be a *zero-sum game*. Otherwise it is said to be *non-zero sum game*.

Strategy The strategy for a player is the list of all possible actions (moves, decision alternatives or courses of action) that are likely to be adopted by him for every payoff (outcome). It is assumed that the players are aware of the rules of the game governing their decision alternatives (or strategies). The outcome resulting from a particular strategy is also known to the players in advance and is expressed in terms of numerical values (e.g. money, per cent of market share or utility).

The particular strategy that optimizes a player's gains or losses, without knowing the competitor's strategies, is called *optimal strategy*. The expected outcome, when players use their optimal strategy, is called *value of the game*.

Generally, the following two types of strategies are followed by players in a game:

- Pure Strategy** A particular strategy that a player chooses to play again and again regardless of other player's strategy, is referred as *pure strategy*. The objective of the players is to maximize their gains or minimize their losses.
- Mixed Strategy** A set of strategies that a player chooses on a particular move of the game with some fixed probability are called *mixed strategies*. Thus, there is a probabilistic situation and objective of the each player is to maximize expected gain or to minimize expected loss by making the choice among pure strategies with fixed probabilities.

Mathematically, if p_j ($j = 1, 2, \dots, n$) is the probability associated with a pure strategy j to be chosen by a player at any point in time during the game, then the set S of n non-negative real numbers

Strategy in a course of action that a player adopts for every payoff (outcome).

Pure strategy is the only course of action that is always chosen by a player.

(probabilities) whose sum is unity associated with pure strategies of the player is written as: $S = \{p_1, p_2, \dots, p_n\}$ where $p_1 + p_2 + \dots + p_n = 1$ and $p_j \geq 0$ of all j .

Remark If a particular $p_j = 1$ ($j = 1, 2, \dots, n$) and all others are zero, the player is said to select pure strategy j . A flow chart of using game theory approach to solve a problem is shown in Fig. 12.1.

Mixed strategy are courses of action that are to be selected on a particular occasion with some fixed probability.

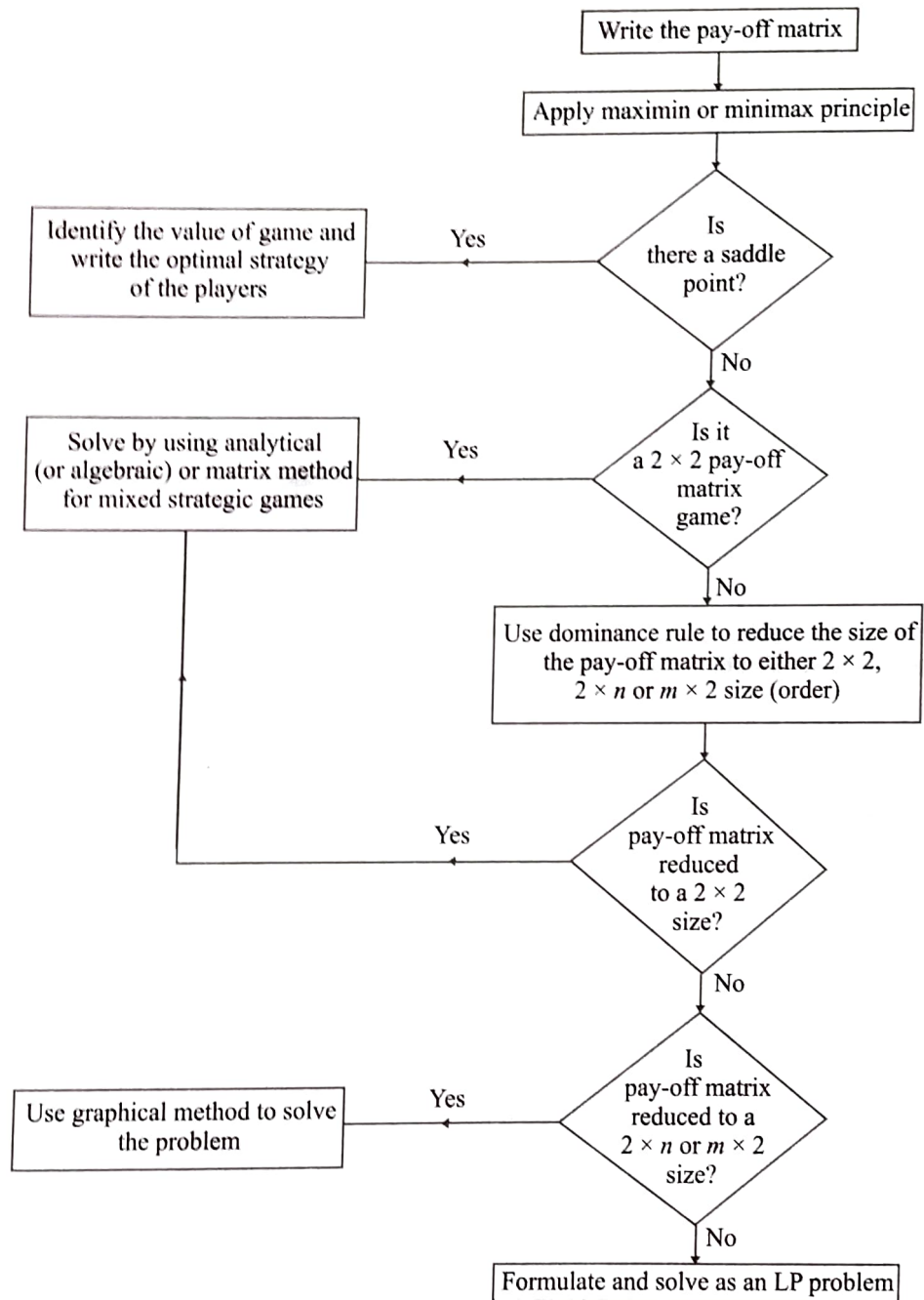


Fig. 12.1
Flow Chart of
Game Theory
Approach

12.2 TWO-PERSON ZERO-SUM GAMES

A game with only two players, say A and B , is called a *two-person zero-sum game*, only if one player's gain is equal to the loss of other player, so that total sum is zero.

Payoff matrix The payoffs (a quantitative measure of satisfaction that a player gets at the end of the play) in terms of gains or losses, when players select their particular strategies (courses of action), can be represented in the form of a matrix, called the payoff matrix. Since the game is zero-sum, the gain of one player is equal to the loss of other and vice versa. In other words, one player's payoff table would contain the same amounts in payoff table of other player, with the sign changed. Thus, it is sufficient to construct a payoff table only for one of the players.

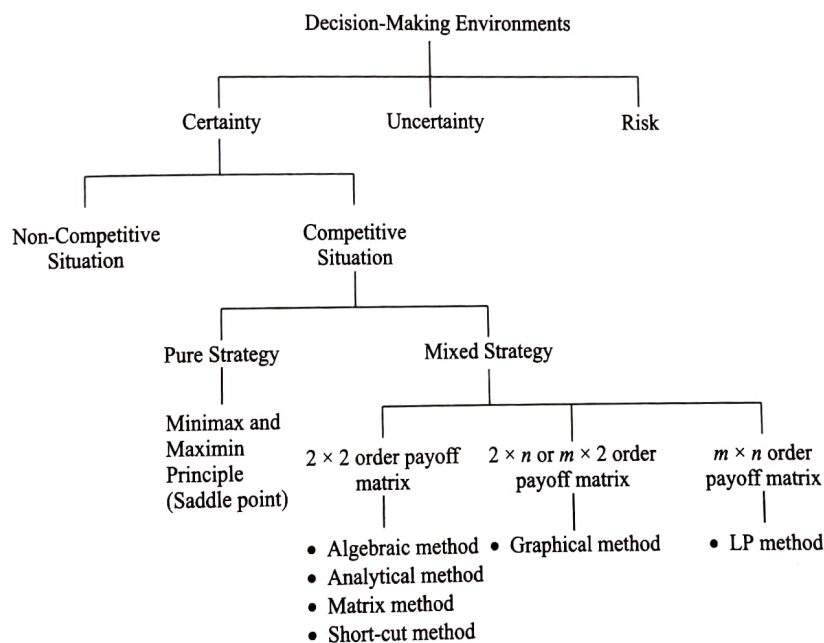
If player A has m strategies represented by the letters: $A_1, A_2, \dots, A_m$ and player B has n strategies represented by the letters: $B_1, B_2, \dots, B_n$. The numbers m and n need not be equal. The total number of possible outcomes is therefore $m \times n$. It is assumed that each player not only knows his own list of possible strategies but also of his competitor. For convenience, it is assumed that player A is always a gainer whereas player B a loser. Let a_{ij} be the payoff that player A gains from player B if player A chooses strategy i and player B chooses strategy j . Then the payoff matrix is shown in the Table 12.1.

Player A's Strategies	Player B's Strategies			
	B_1	B_2	$\dots$	B_n
A_1	a_{11}	a_{12}	$\dots$	a_{1n}
A_2	a_{21}	a_{22}	$\dots$	a_{2n}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
A_m	a_{m1}	a_{m2}	$\dots$	a_{mn}

Table 12.1
Payoff Matrix

Since player A is assumed to be the gainer, therefore he wishes to gain as large a payoff a_{ij} as possible, player B on the other hand would do his best to reach as small a value of a_{ij} as possible. Of course, the gain to player B and loss to A must be $-a_{ij}$.

Various methods discussed in this chapter to find value of the game under decision-making environment of certainty are as follows:



Assumptions of the game

1. Each player has available to him a finite number of possible strategies (courses of action). The list may not be the same for each player.
2. Players act rationally and intelligently.
3. List of strategies of each player and the amount of gain or loss on an individual's choice of strategy is known to each player in advance.
4. One player attempts to maximize gains and the other attempts to minimize losses.
5. Both players make their decisions individually, prior to the play, without direct communication between them.
6. Both players select and announce their strategies simultaneously so that neither player has an advantage resulting from direct knowledge of the other player's decision.
7. The payoff is fixed and determined in advance.

12.3 PURE STRATEGIES (MINIMAX AND MAXIMIN PRINCIPLES): GAMES WITH SADDLE POINT

Maximin principle maximize the player's minimum gains.

The selection of an optimal strategy by each player, without the knowledge of the competitor's strategy, is the basic problem of playing games. Since the payoffs for either player provides all the essential information, therefore, only one player's payoff table is required to evaluate the decisions. By convention, the payoff table for the player whose strategies are represented by rows (say player A) is constructed. The objective of the study is to know how these players must select their respective strategies so that they are able to optimize their payoff. Such a decision-making criterion is referred to as the *minimax-maximin principle*. Such principle in pure strategies game always leads to the best possible selection of a strategy for both players.

Maximin principle For player A the minimum value in each row represents the least gain (payoff) to him, if he chooses his particular strategy. These are written in the matrix by row minima. He will then select the strategy that gives the largest gain among the row minimum values. This choice of player A is called the *maximin principle*, and the corresponding gain is called the *maximin value of the game*.

Minimax principle For player B , who is assumed to be the looser, the maximum value in each column represents the maximum loss to him, if he chooses his particular strategy. These are written in the payoff matrix by column maxima. He will then select the strategy that gives the minimum loss among the column maximum values. This choice of player B is called the *minimax principle*, and the corresponding loss is the *minimax value of the game*.

Optimal strategy A course of action that puts any player in the most preferred position, irrespective of the course of action his competitor(s) adopt, is called as optimal strategy. In other words, if the maximin value equals the minimax value, then the game is said to have a *saddle (equilibrium) point* and the corresponding strategies are called *optimal strategies*.

Value of the game This is the expected payoff at the end of the game, when each player uses his optimal strategy, i.e. the amount of payoff, V , at an equilibrium point. A game may have more than one saddle points. A game with no saddle point is solved by choosing strategies with fixed probabilities.

- Remarks**
1. The value of the game, in general, satisfies the equation: maximin value $\leq V \leq$ minimax value.
 2. A game is said to be a *fair game* if the lower (maximin) and upper (minimax) values of the game are equal and both equals zero.
 3. A game is said to be *strictly determinable* if the lower (maximin) and upper (minimax) values of the game are equal and both equal the value of the game.

12.3.1 Rules to Determine Saddle Point

The reader is advised to follow the following three steps, in this order, to determine the saddle point in the payoff matrix.

1. Select the minimum (lowest) element in each row of the payoff matrix and write them under 'row minima' heading. Then, select the largest element among these elements and enclose it in a rectangle, $\boxed{}$.
2. Select the maximum (largest) element in each column of the payoff matrix and write them under 'column maxima' heading. Then select the lowest element among these elements and enclose it in a circle, $\bigcirc$.
3. Find out the element(s) that is same in the circle as the well as rectangle and mark the position of such element(s) in the matrix. This element represents the value of the game and is called the saddle (or equilibrium) point.

Example 12.1 For the game with payoff matrix:

Player A	Player B		
	B_1	B_2	B_3
A_1	-1	2	-2
A_2	6	4	-6

determine the optimal strategies for players A and B . Also determine the value of game. Is this game (i) fair? (ii) strictly determinable?

Solution In this example, gains to player A or losses to player B are represented by the positive quantities, whereas, losses to A and gains to B are represented by negative quantities. It is assumed that A wants to

Value of the game is the expected gain or loss in a game when a game is played a large number of times.

Saddle point is the payoff value that represents both minimax and maximin value of the game.

maximize his minimum gains from B . Since the payoffs given in the matrix are what A receives, therefore, he is concerned with the quantities that represent the row minimums. Now A can do no worse than receive one of these values. The best of these values occurs when he chooses strategy A_1 . This choice provides a payoff of -2 to A when B chooses strategy B_3 . This refers to A 's choice of A_1 as his maximum payoff strategy because this row contains the maximum of A 's minimum possible payoffs from his competitor B .

Player A	Player B			Row minimum
	B_1	B_2	B_3	
A_1	-1	2	-2	$-2 \leftarrow \text{Maximin}$
A_2	6	4	-6	-6
Column maximum	6	4	$-2 \leftarrow \text{Minimax}$	

Table 12.2

Similarly, it is assumed that B wants to minimize his losses and wishes that his losses to A be as small as possible. The column maximums also represent the greatest payments B might have to make to A . The smallest of these losses is -2 , which occurs when A chooses his course of action, A_1 and B chooses his course of action, B_3 . This choice of B_3 by B is his minimax loss strategy because the amount of this column is the minimum of the maximum possible losses.

In Table 12.2, the quantity -2 in row A_1 and column B_3 is enclosed both in the box and the circle. That is, it is both the minimum of the column maxima and the maximum of the row minima. This value is referred to as *saddle point*.

The payoff amount in the saddle-point position is also called *value of the game*. For this game, value of the game is, $V = -2$, for player A . The value of game is always expressed from the point of view of the player whose strategies are listed in the rows.

The game is strictly determinable. Also since the value of the game is not zero, the game is not fair.

Example 12.2 A company management and the labour union are negotiating a new three year settlement. Each of these has 4 strategies:

- I : Hard and aggressive bargaining II : Reasoning and logical approach
 III : Legalistic strategy IV : Conciliatory approach

The costs to the company are given for every pair of strategy choice.

Union Strategies	Company Strategies			
	I	II	III	IV
I	20	15	12	35
II	25	14	8	10
III	40	2	10	5
IV	-5	4	11	0

What strategy will the two sides adopt? Also determine the value of the game.

Solution Applying the rule of finding out the saddle point, we obtain the saddle point that is enclosed both in a circle and a rectangle, as shown in Table 12.3.

Union Strategies	Company Strategies				Row minimum
	I	II	III	IV	
I	20	15	12	35	12 $\leftarrow \text{Maximin}$
II	25	14	8	10	8
III	40	2	10	5	2
IV	-5	4	11	0	-5
Column maximum	40	15	12 $\uparrow \text{Minimax}$	35	

Table 12.3

As shown in Table 12.3, since $\text{Maximin} = \text{Minimax} = \text{Value of game} = 12$, therefore the company will always adopt strategy III – Legalistic strategy and union will always adopt strategy I – Hard and aggressive bargaining.

Example 12.3 Find the range of values of p and q that will render the entry $(2, 2)$ a saddle point for the game:

		Player B		
Player A		B_1	B_2	B_3
A_1		2	4	5
A_2		10	7	q
A_3		4	p	6

Solution First ignore the values of p and q in the payoff matrix, and then determine the maximin and minimax values in the usual manner, as shown in Table 12.4.

		Player B			
Player A		B_1	B_2	B_3	Row minimum
A_1		2	4	5	2
A_2		10	7	q	7 ← Maximin
A_3		4	p	6	4
Column maximum		10	7	6	
				↑ Minimax	

Table 12.4

As shown in Table 12.4, since there exists no unique saddle point, therefore, the saddle point will exist at the position $(2, 2)$ only when $p \leq 7$ and $q > 7$.

Example 12.4 For what value of λ , the game with following pay-off matrix is strictly determinable?

		Player B		
Player A		B_1	B_2	B_3
A_1		λ	6	2
A_2		-1	λ	-7
A_3		-2	4	λ

Solution First, ignoring the value of λ , determine the maximin and minimax values of the pay-off matrix, as shown below:

		Player B			
		B_1	B_2	B_3	Row minimum
A_1		λ	6	2	2 ← Maximin
A_2		-1	λ	-7	-7
A_3		-2	4	λ	-2
Column maximum		-1	6	2	
		↑ Minimax			

Since saddle point in the above table is not unique, the value of the game lies between -1 and 2 , i.e. $-1 \leq V \leq 2$. For strictly determinable game, we must have $-1 \leq \lambda \leq 2$.

CONCEPTUAL QUESTIONS A

- Define: (i) competitive game, (ii) payoff matrix, (iii) pure and mixed strategies, (iv) saddle point, (v) optimal strategies, and (vi) rectangular (or two-person zero-sum) game.
- Explain: Minimax and Maximin principle used in the theory of games.
- What is a game in game theory? What are the properties of a game? Explain the 'best strategy' on the basis of minimax criterion of optimality. [Delhi Univ., MBA, 2006]
- Explain the two-person zero-sum game, giving a suitable example.
- Define 'saddle point'. Is it necessary that a game should always possess a saddle point? [Agra MCA, 2000]
- Let $A = [a_{ij}]$ be the payoff matrix for a two-person zero-sum game. If V denotes the minimax value of the game, then show that $V \geq \bar{V}$. [Meerut MSc (Stat.), 2002]
- Define: (i) Competitive game; (ii) Pure strategies; (iii) Mixed strategies (iv) Two-person zero-sum (or rectangular) game. (v) Payoff matrix.
- State the major limitations of the game theory. [Delhi Univ., MCom, 2000]